The Association Between U.S. Patients' Household Income & Medication Experiences

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Objectives

Examine the association between U.S patients' household income and their medication experiences in terms of:

1) Medication-Related Burden

2) Medication-Related Beliefs

3) Medication-Taking Practices



Background

- 80% of the US adult (18 and older) population takes at least one medication per week [1].
- A typical person will take 14,000 pills in his/her lifetime [2].
- About 22% of older people (65 years and older) and about 31% of low-income older adults experience medication underuse because of medication cost [3].
- One out of five private insurance users (popular among higher income patients) also report cost-related medication underuse [3].
- There is a lack of research on the association between U.S patients' household income and their medication experiences.

Methods

- The data source was collected from the **2015 National Consumer Survey of the Medication Experience and Pharmacists' Roles.**
- A cross-sectional, self-administered, on-line survey was used for collecting and analyzing data using the Dillman Tailored Design method and through the use of technical assistance from Qualtrics Panels.
- The survey was conducted April 28, to June 22, 2015 in U.S.
- Based on work by Mohamed, Moles, and Chen, three variables were used to describe the patient medication experience: (1) medication related burden, (2) medication related beliefs, and (3) medication taking practices [4].
- Specific items were created within each of these categories and statistical significance between groups of respondents was set at a p-value less than 0.05 (Chi-square analysis).
- Data was transferred from Qualtric Panels Excel Spreadsheet into an IBM SPSS data file. Descriptive Statistics (frequencies, mean, median, mode, range) were computed.

Distribution of Respondents

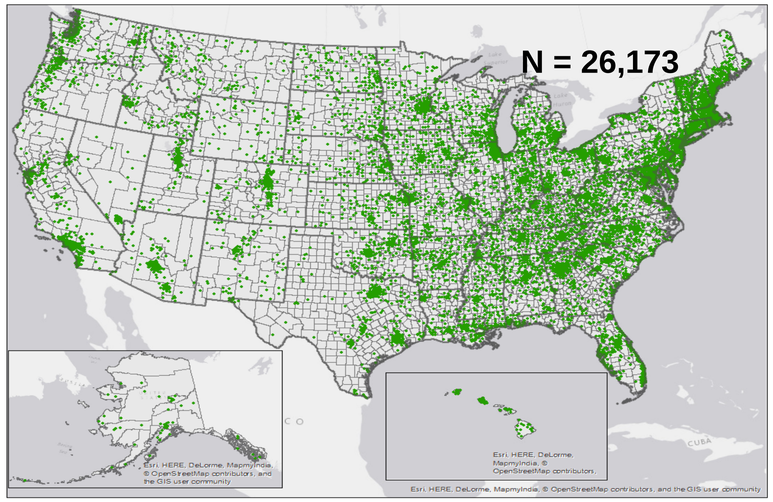
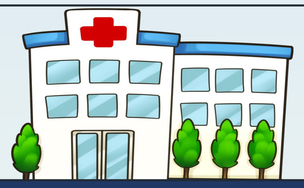
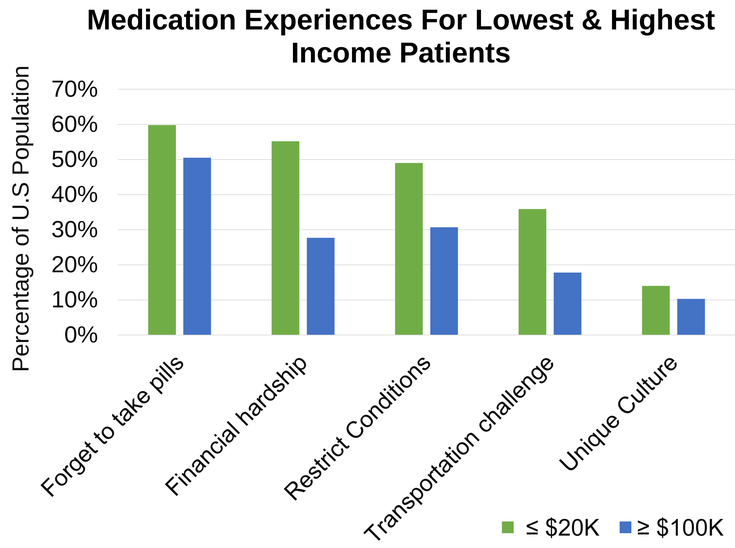


Figure 1 shows the geographic distribution within states was similar to that of the U.S. population; post-stratification weighting would be applied for estimates for the overall U.S. population.



Results



Graph 1 compares specific scenario of medication experiences between highest household income (≥ $100k) & lowest household income (≤ $20k) patients.

Results

Medication Experiences	≤ $20k	$21 - $40k	$41- $60k	$61- $80k	$81- $100k	≥ $100k
Financial Hardship	55.2%	53.2%	47.7%	42.3%	38.3%	27.7%
Transportation Challenge	35.9%	25.1%	22.0%	21.0%	21.1%	17.8%
Unique Culture	14.0%	11.8%	11.2%	11.7%	13.3%	10.3%
Forget to take medications	59.8%	54.7%	53.5%	54.5%	55.0%	50.5%
Restricted life conditions	49.0%	41.5%	36.7%	34.8%	33.9%	30.7%
Professional relationship w/ a pharmacist	27.4%	27.1%	27.0%	26.0%	26.2%	23.8%
Use MTM	2.7%	2.1%	2.8%	3.3%	4.6%	2.5%

Table 1: Summary of the Key Findings. P-values for comparisons among the income categories were statistically significant (p-values < 0.05).

Recommendations

- Only 3% of the U.S. population reported using Medication Therapy Management (MTM) services. MTM allows patients and pharmacist to discuss concerns regarding their medication experiences. Pharmacists can come up with the most effective medication plans and treatments according to individual patient needs. Efforts to increase the use of MTM are warranted.
- Other recommendations include:
  - > Develop discounts for prescription drugs.
  - > Help patients with transportation needs.
  - > Support of minority students who have the potential to be healthcare professionals.
  - > Cultural competency training.
  - > More study on medication experiences.

Conclusions

- Lower household income patients are more prone to have medication-related burden, negative medication beliefs, and less likely to have good medication taking practices.
- Since low MTM usage is linked to undesirable medication experience, higher access will be the best solution.